

CH NO 06:

Probability Distribution

A probability Distribution describe how the probabilities of different outcomes are distributed in a random experiment.

→ It is a mathematical function that gives the probability of occurrence of different possible outcomes.

Types:

1- Discrete Probability Distribution:

→ (Used for countable outcomes)

e.g

Binomial Distribution etc.

2- Continuous P.D:

→ (for continuous outcomes)

e.g

Normal Distribution etc.

Binomial Distribution

Conditions for B.D:

- i- Finite number of trials
- ii- Independent event (with replacement)
- iii- Probability of every event remain same-
- iv- Dual in nature
 - ↳ Success
 - ↳ Failure

$$P(X=x) = \binom{n}{x} p^x q^{n-x}$$

n = No of trial

p = Probability of success in trial

q = Probability of failure in trial

x = No of success ($x = 0, 1, 2, \dots, n$)

Example no 05

 $n = 6$ tosses $x = 4$ Heads

Sol.,

$$P = \frac{\{H\}}{\{H, T\}} = \frac{1}{2} = \boxed{0.5}$$

As we know that:

$$P + q = 1$$

$$0.5 + q = 1$$

$$q = 1 - 0.5$$

$$\boxed{q = 0.5}$$

Now,

$$P(X=x) = \binom{n}{x} P^x q^{n-x}$$

$$P(X=4) = {}^6C_4 (0.5)^4 (0.5)^{6-4}$$

$$= (15)(0.0625)(0.5)^2$$

$$= (15)(0.0625)(0.25)$$

$$\boxed{P(X=4) = 0.2343}$$

Range $0, 1, 2, \dots, \infty$

(M)(T)(W)(T)(F)(S)

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Example:

$$p = \frac{1}{4}$$

$$n = 5$$

Sol.

Formula:

$$P(X=x) = {}^nC_x p^x q^{n-x}$$

For q :

$$p + q = 1$$

$$\frac{1}{4} + q = 1$$

$$q = 1 - \frac{1}{4}$$

$$= \frac{4-1}{4}$$

$$\boxed{q = \frac{3}{4}}$$

i) $P(x=0)$

$$P(X=0) = {}^5C_0 \left(\frac{1}{4}\right)^0 \left(\frac{3}{4}\right)^{5-0}$$

$$= (1)(1)\left(\frac{3}{4}\right)^5$$

$$= (1)(1)\left(\frac{243}{1024}\right)$$

$$P(X=0) = 0.2373$$

ii) $P(X \leq 3)$

$$P = \frac{1}{4}$$

$$n = 5$$

$$X = 0, 1, 2, 3$$

Sol: -

$$P + q = 1$$

$$\frac{1}{4} + q = 1$$

$$q = 1 - \frac{1}{4}$$

$$q = \frac{4-1}{4}$$

$$q = \frac{3}{4}$$

Now,

For $X=0$

$$P(X=0) = 0.2373$$

For $x=1$

$$P(X=1) = {}^5C_1 \left(\frac{1}{4}\right)^1 \left(\frac{3}{4}\right)^{5-1}$$

$$= (5) \left(\frac{1}{4}\right) \left(\frac{3}{4}\right)^4$$

$$= (5) \left(\frac{1}{4}\right) \left(\frac{81}{256}\right)$$

$$= \frac{648}{1024}$$

$$= \boxed{0.6328}$$

For $x=2$

$$P(X=2) = {}^5C_2 \left(\frac{1}{4}\right)^2 \left(\frac{3}{4}\right)^{5-2}$$

$$= (10) \left(\frac{1}{16}\right) \left(\frac{3}{4}\right)^3$$

$$= (10) \left(\frac{1}{16}\right) \left(\frac{27}{64}\right)$$

$$= \frac{270}{1024}$$

$$= \boxed{0.2636}$$

For $x=3$

$$P(X=3) = {}^5C_3 \left(\frac{1}{4}\right)^3 \left(\frac{3}{4}\right)^{5-3}$$

$$= (10) \left(\frac{1}{64}\right) \left(\frac{3}{4}\right)^2 = (10) \left(\frac{1}{64}\right) \left(\frac{9}{16}\right)$$

$$= \frac{90}{1024}$$

$$= \boxed{0.0878}$$

) X

Questions related to Binomial Distribution

Q.1	A fair coin is tossed 4 times. Find the probabilities of obtaining various numbers of heads.
Q.2	A fair coin is tossed 5 times. What is the probability of getting? (i) Exactly 3 heads (ii) At least 3 heads (iii) At the most two heads
Q.3	In a Binomial distribution $n = 20$ and $p = 3/5$. Find the mean, variance, and standard deviation of the distribution.
Q.4	In a Binomial distribution, the mean and standard deviation were found to be 38 and 5.6 respectively. Find p and n .
Q.5	Given the probability of passing an examination is 0.75. What is the probability of: (i) Passing at least two examinations if you take six? (ii) Failing at least two examinations if you take four?
Q.6	The accident of an occupational disease in an industry is such that the workers have 20% chance of suffering from it. What is the probability that out of 6 workman: (i) Not more than 2 will catch the disease? (ii) 4 or more will catch the disease?
Q.7	If X is the number of successes with probability of success as $1/4$ in each of 5 independent trial. Find: (i) $P(X = 0)$ (ii) $P(X \leq 3)$
Q.8	If 5 true dice are thrown once, determine the probability of getting 0,1,2,3,4,5 sixes. Find the mean and variance of the probability distribution so obtained.

Binomial Distribution

Example 1:

$$n = 4$$

$$4 \text{ tossed} \Rightarrow n = 4$$

$$2^n = 2^4 = 16$$

Let x will be number of Heads
when a coin tossed 4 times.

$$x = 0, 1, 2, 3, 4$$

And P is the probability of success for
getting Heads in each tossed:

$$P = \frac{\{H\}}{\{H, T\}} = \frac{1}{2} = 0.5$$

$$P + q = 1$$

$$0.5 + q = 1$$

$$q = 1 - 0.5$$

$$q = 0.5$$

Sol.

$$P(X=x) = {}^nC_x P^x q^{n-x}$$

Number of Heads obtained:

For 0 Heads:

$$\begin{aligned} P(X=0) &= {}^4C_0 (0.5)^0 (0.5)^{4-0} \\ &= (1)(1)(0.5)^4 \end{aligned}$$

$$P(X=0) = 0.0625$$

For 1 Heads:

$$\begin{aligned} P(X=1) &= {}^4C_1 (0.5)^1 (0.5)^{4-1} \\ &= (4)(0.5)(0.5)^3 = (4)(0.5)(0.125) \\ &= 0.25 \end{aligned}$$

For 2 Heads:

$$P(X=2) = {}^4C_2 (0.5)^2 (0.5)^{4-2}$$

$$= (6)(0.25)(0.5)^2$$

$$= (6)(0.25)(0.25)$$

$$P(X=2) = 0.375$$

For 3 Heads:

$$P(X=3) = {}^4C_3 (0.5)^3 (0.5)^{4-3}$$

$$= (4)(0.125)(0.5)^1$$

$$= (4)(0.125)(0.5)$$

$$P(X=3) = 0.25$$

For 4 Heads:

$$P(X=4) = {}^4C_4 (0.5)^4 (0.5)^{4-4}$$

$$= (1)(0.0625)(0.5)^0$$

$$P(X=4) = 0.0625$$

x _____ x

Example 3:

Given:

$$n = 20$$

$$p = \frac{3}{5}$$

Find:

$$\text{Mean} = ?$$

$$\text{Variance} = ?$$

$$\text{S.D} = ?$$

Sol.

① Mean of the distribution:

$$\begin{aligned}\mu &= n \cdot P \\ &= (20) \left(\frac{3}{5} \right) \\ &= (4)(3)\end{aligned}$$

$$\boxed{\mu = 12}$$

② Standard deviation of the distribution:

$$\begin{aligned}\sigma^0 &= \sqrt{n \cdot p (1-p)} & (\because \text{S.D} = \sqrt{\text{Variance}}) \\ &= \sqrt{(20) \left(\frac{3}{5} \right) \left(1 - \frac{3}{5} \right)}\end{aligned}$$

$$= \sqrt{(12) \left(\frac{5-3}{5} \right)}$$

$$= \sqrt{(12) \left(\frac{2}{5} \right)}$$

$$= \sqrt{\frac{24}{5}}$$

$$\boxed{\sigma = 2.19}$$

② Variance of the distribution:

$$\text{Variance } (\sigma^2) = n \cdot P \cdot (1-P)$$

$$= {}^{(20)} \left(\frac{3}{5} \right) \left(1 - \frac{3}{5} \right)$$

$$= {}^{(12)} \left(\frac{5-3}{5} \right)$$

$$\boxed{6^2 = \frac{24}{5}}$$

$$x \text{ ————— } x$$

Example 2:

$$\boxed{n = 5}$$

$$2^n = 2^5 = 32$$

Let $\Rightarrow x$ will be numbers of (Exactly 3 Heads)

$\Rightarrow y$ will be numbers of (At least 3 Heads)

$\Rightarrow z$ will be numbers of (At the most two heads)

P is the probability of success for getting Heads in each toss.

Sol:

① Exactly 3 Heads

$$\boxed{x = 3}$$

$$P(X=x) = {}^nC_x p^x q^{n-x}$$

$$P(X=3) = {}^5C_3 (0.5)^3 (0.5)^{5-3}$$

$$= (10)(0.125)(0.5)^2$$

$$= (10)(0.125)(0.25)$$

$$\boxed{P(X=3) = 0.3125}$$

$$P = \frac{\{H\}}{\{H, T\}} = \frac{1}{2} = \boxed{0.5}$$

$$\therefore P + q = 1$$

$$q = 1 - P$$

$$= 1 - 0.5$$

$$\boxed{q = 0.5}$$

② At least 3 Heads $y \geq 3$

$$P(X=y) = {}^nC_y p^y q^{n-y}$$

For $y=3$:

$$\begin{aligned} P(Y=3) &= {}^5C_3 (0.5)^3 (0.5)^{5-3} \\ &= (10)(0.125)(0.25) \end{aligned}$$

$$P(Y=3) = 0.3125$$

For $y=4$

$$\begin{aligned} P(Y=4) &= {}^5C_4 (0.5)^4 (0.5)^{5-4} \\ &= (5)(0.0625)(0.5) \end{aligned}$$

$$P(Y=4) = 0.15625$$

For $y=5$

$$\begin{aligned} P(Y=5) &= {}^5C_5 (0.5)^5 (0.5)^{5-5} \\ &= (1)(0.03125)(0.5)^0 \\ &= (1)(0.03125)(1) \end{aligned}$$

$$P(Y=5) = 0.03125$$

Now; Probability of getting
At least 3 Heads ($y \geq 3$)

$$\begin{aligned} P(y \geq 3) &= P(Y=3) + P(Y=4) + P(Y=5) \\ &= 0.3125 + 0.15625 + 0.03125 \end{aligned}$$

$P(y \geq 3) = 0.5$

③ At the most two heads ($Z \leq 2$)

$$P(Z=z) = {}^nC_z p^z q^{n-z}$$

For $z=2$

$$P(Z=2) = {}^5C_2 (0.5)^2 (0.5)^{5-2}$$

$$= (10) (0.25) (0.5)^3$$

$$= (10) (0.25) (0.125)$$

$$P(Z=2) = 0.3125$$

For $z=1$

$$P(Z=1) = {}^5C_1 (0.5)^1 (0.5)^{5-1}$$

$$= (5) (0.5) (0.5)^4$$

$$= (5) (0.5) (0.0625)$$

$$P(Z=1) = 0.15625$$

For $z=0$

$$P(Z=0) = {}^5C_0 (0.5)^0 (0.5)^{5-0}$$

$$= (1) (1) (0.03125)$$

$$= 0.03125$$

$$P(Z=0) = 0.03125$$

For $z=5$

$$P(Z=5) = {}^5C_5 (0.5)^5 (0.5)^{5-5}$$

$$= (1) (0.3125) (0.5)^0$$

$$P(Z=5) = 0.3125$$

Now, probability of At the most two heads ($Z \leq 2$) is:

$$P(Z \leq 2) = P(Z=0) + P(Z=1) + P(Z=2)$$

$$= 0.03125 + 0.15625 + 0.3125$$

$$P(Z \leq 2) = 0.5$$

x ————— x

Example 4:

Given:

$$\text{Mean} = 38$$

$$\text{Standard deviation} = 5.6$$

Find

$$P = ?$$

$$n = ?$$

Sol.

$$\text{Mean } (\mu) = n \cdot P = 38 \text{ ——— ①}$$

and

$$\text{S.D } (\sigma) = \sqrt{n \cdot P \cdot (1-P)} = 5.6$$

Squaring on b.s;

$$(\sqrt{n \cdot P \cdot (1-P)})^2 = (5.6)^2$$

$$n \cdot P \cdot (1-P) = 31.36$$

Put value of n.P from ①:

$$(38)(1-P) = 31.36$$

$$1-P = \frac{31.36}{38}$$

$$1 - P = 0.8252$$

$$\Rightarrow P = 1 - 0.8252$$

$$P = 0.1748$$

put value of p in (1):

$$n \cdot P = 38$$

$$n = \frac{38}{P}$$

$$n = \frac{38}{0.1748}$$

$$n = 217.5$$

Example 5:

Given: $P = 0.75$

Let x will be number of passed exams out of 6.

$$P + q = 1$$

$$q = 1 - P = 1 - 0.75$$

$$q = 0.25$$

① Probability of passing at least two exams if you take six (6): $(X \geq 2)$

Here,

$$n = 6$$

$$P(X=x) = {}^n C_x P^x q^{n-x}$$

For $x=2$

$$\begin{aligned} P(X=2) &= {}^6 C_2 (0.75)^2 (0.25)^{6-2} \\ &= (15) (0.5625) (0.25)^4 \end{aligned}$$

$$= (15)(0.5625)(0.003906)$$

$$= 0.03296$$

For $x=1$

$$P(X=1) = {}^6C_1 (0.75)^1 (0.25)^{6-1}$$

$$= (6)(0.75)(0.25)^5$$

$$= (6)(0.75)(0.0009766)$$

$$P(X=1) = 0.004395$$

For $x=0$

$$P(X=0) = {}^6C_0 (0.75)^0 (0.25)^{6-0}$$

$$= (1)(1)(0.0002442)$$

$$= 0.0002442$$

Now,

$$P(X \geq 2) = P(X=0) + P(X=1) + P(X=2)$$

$$= 0.0002442 + 0.004395 + 0.03296$$

$$= 0.03760$$

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Example 07:

Given:

$$P = \frac{1}{4}$$

$$n = 5$$

Find

i) $P(X=0)$

ii) $P(X \leq 3)$

Sol: $P(X=x) = {}^nC_x P^x q^{n-x}$

As we know that:

$$P + q = 1$$

$$q = 1 - P$$

$$= 1 - \frac{1}{4}$$

$$= \frac{4-1}{4}$$

$$q = \frac{3}{4}$$

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i) Case-I ($P(x=0)$)

$$P(X=0) = {}^5C_0 \left(\frac{1}{4}\right)^0 \left(\frac{3}{4}\right)^{5-0}$$

$$= (1)(1) \left(\frac{243}{1024}\right)$$

$$P(X=0) = 0.2373$$

ii) Case-II ($P(X \leq 3)$)At $x=3$

$$P(X=3) = {}^5C_3 \left(\frac{1}{4}\right)^3 \left(\frac{3}{4}\right)^{5-3}$$

$$= (10) \left(\frac{1}{64}\right) \left(\frac{3}{4}\right)^2$$

$$= \left(\frac{10}{64}\right) \left(\frac{9}{16}\right) = \frac{90}{1024}$$

$$P(X=3) = 0.08790$$

At $x=2$

$$P(X=2) = {}^5C_2 \left(\frac{1}{4}\right)^2 \left(\frac{3}{4}\right)^{5-2}$$

$$= (10) \left(\frac{1}{16}\right) \left(\frac{3}{4}\right)^3$$

$$= \left(\frac{10}{16}\right) \left(\frac{27}{64}\right) = \frac{270}{1024}$$

$$P(X=2) = 0.2637$$

At $x=1$

$$P(X=1) = {}^5C_1 \left(\frac{1}{4}\right)^1 \left(\frac{3}{4}\right)^{5-1}$$

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$$= (5) \left(\frac{1}{4}\right) \left(\frac{3}{4}\right)^4$$

$$= \left(\frac{5}{4}\right) \left(\frac{81}{256}\right) = \frac{405}{1024}$$

$$P(X=1) = 0.3956$$

At $x=0$

$$P(X=0) = {}^5C_0 \left(\frac{1}{4}\right)^0 \left(\frac{3}{4}\right)^{5-0}$$

$$= (1)(1) \left(\frac{243}{1024}\right)$$

$$P(X=0) = 0.2373$$

Now,

$$P(X \leq 3) = P(X=0) + P(X=1) + P(X=2) + P(X=3)$$

$$= 0.2373 + 0.3956 + 0.2637 + 0.08790$$

$$P(X \leq 3) = 0.9845$$

x _____ x

Example 8:Given:

$$n=5$$

P is the probability of success
(getting a specific number a die):

$$P = \frac{1}{6} = 0.1666$$

As we know that:

(M) (T) (W) (T) (F) (S)

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$$p + q = 1$$

$$q = 1 - p$$

$$= 1 - 0.1666$$

$$q = 0.8333$$

And, ① Probability of getting sixes:

$$P(X=x) = {}^nC_x p^x q^{n-x}$$

For getting 0 six:

$$P(X=0) = {}^5C_0 (0.1666)^0 (0.8333)^{5-0}$$

$$= (1)(1)(0.8333)^5$$

$$P(X=0) = 0.4018$$

For getting 1 six:

$$P(X=1) = {}^5C_1 (0.1666)^1 (0.8333)^{5-1}$$

$$= (5)(0.1666)(0.8333)^4$$

$$= (5)(0.1666)(0.4822)$$

$$P(X=1) = 0.4017$$

For getting 2 six:

$$P(X=2) = {}^5C_2 (0.1666)^2 (0.8333)^{5-2}$$

$$= (10)(0.02776)(0.8333)^3$$

$$= (10)(0.02776)(0.5787)$$

$$P(X=2) = 0.1606$$

For getting 3 six

$$P(X=3) = {}^5C_3 (0.1666)^3 (0.8333)^{5-3}$$

$$= (10)(0.004624)(0.8333)^2$$

$$= (10)(0.004624)(0.6944)$$

(M) (T) (W) (T) (F) (S)

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$$P(X=3) = 0.03211$$

For getting 4 six

$$P(X=4) = {}^5C_4 (0.1666)^4 (0.8333)^{5-4}$$

$$= (5)(0.0007704)(0.8333)$$

$$P(X=4) = 0.003210$$

For getting 5 six

$$P(X=5) = {}^5C_5 (0.1666)^5 (0.8333)^{5-5}$$

$$= (1)(0.0001283)(1)$$

$$P(X=5) = 0.0001283$$

② Mean:

$$\mu = n \cdot p$$

$$= (5)(0.1667)$$

$$\mu = 0.8335$$

③ Variance:

$$\sigma^2 = n \cdot p \cdot (1 - p)$$

$$= (5)(0.1667)(1 - 0.1667)$$

$$= (5)(0.1667)(0.8333)$$

$$\sigma^2 = 0.6945$$

x ————— x

Q.1	The score made by candidate in a certain test are normally distributed with mean 500 and standard deviation 100. What percent of the candidates received scores: (i) Greater than 700 (ii) Less than 400 (iii) Between 400 and 600 (iv) Which differ from mean by more than 150
Q.2	A department store salesclerk knows that the number of sales she will make on a business day is a random variable having approximately a normal distribution with mean = 25 and standard deviation = 4. Find the probabilities that during a business day the salesclerk will make: (i) More than 26 sales (ii) Fewer than 20 sales (iii) At least 25 or at most 25
Q.3	In a normal distribution with mean = 20 and standard deviation = 5. Find the areas: (i) Between 20 and 30 (ii) Less than 25 (iii) More than 30 (iv) Between 12 and 18 (v) Between 30 and 40
Q.4	The heights of freshmen students at a military academy are normally distributed with a mean of 5 feet 10 inches and a standard deviation of 2 inches. (i) What is the proportion of freshmen at the academy who taller than 6 feet 3 inches? (ii) What is the proportion of freshmen who are less than 5 feet 7 inches? (iii) What is the proportion of freshmen between 5 feet 8 inches and 5 feet 0 inches?
Q.5	A machine produces components whose thickness is normally distributed with mean 0.154" and standard deviation 0.002. Components with thicknesses outside the range 0.150" to 0.155" are rejected. What proportion of the production is: (i) Accepted? (ii) Rejected?
Q.6	Let $Y = 2X + 20$ and $X \sim N(15, 4)$. Find (i) $P(Y \leq 40)$ (ii) $P(Y \geq 52)$ (iii) $P(47 \leq Y \leq 55)$
Q.7	Let be a normal random variable with mean = 16 and standard deviation = 5. Determine: (i) $P(\text{between } 11 \text{ and } 21)$ (ii) $P(\text{at least } 26)$ (iii) $P(\text{less than or equal to } 6)$ (iv) $P(\text{at most } 21)$ (v) $P(\text{less than } 10 \text{ or more than } 18)$.

Formulas:

$$P(Z \leq a) = \Phi(a)$$

$$P(Z \geq a) = 1 - \Phi(a)$$

$$P(a \leq Z \leq b) = \Phi(b) - \Phi(a)$$

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Normal Distribution

- Continuous variable

- $(-\infty \leq x \leq +\infty)$

- $x = 0, 1, 2, \dots, n$

$$Z = \frac{x - \mu}{\sigma}$$

i) $P(Z \leq a) = \Phi(a)$

ii) $P(Z \geq b) = 1 - \Phi(b)$

iii) $P(a \leq Z \leq b) = \Phi(b) - \Phi(a)$

Example NO 01:

Given: $\mu = 500$
 $\sigma = 100$

a) Greater than 700

Sol. $P(x \geq 700) = ?$

$$P\left(\frac{x - \mu}{\sigma} \geq \frac{700 - 500}{100}\right)$$

$$P\left(Z \geq \frac{700 - 500}{100}\right)$$

$$P(Z \geq \frac{200}{100})$$

$$P(Z \geq 2) = 1 - \Phi(2) \quad (\because P(Z \geq a) = 1 - \Phi(a))$$

$$= 1 - \Phi(2.00)$$

$$= 1 - 0.97$$

$$P(Z \geq 2) = 0.03$$

b) Less than 400

Sol. $P(x \leq 400) = ?$

$$P\left(\frac{x - \mu}{\sigma} \leq \frac{400 - 500}{100}\right)$$

$$P\left(Z \leq \frac{400 - 500}{100}\right)$$

$$P\left(Z \leq \frac{-100}{100}\right)$$

$$P(Z \leq -1) = \Phi(-1)$$

$$= \Phi(-1.00)$$

$$P(Z \leq -1) = 0.15866$$

c) between 400 and 600

Sol. $P(400 < x \leq 600)$

$$= P\left(\frac{400 - 500}{100} \leq \frac{x - \mu}{\sigma} \leq \frac{600 - 500}{100}\right)$$

$$= P\left(\frac{400 - 500}{100} \leq Z \leq \frac{600 - 500}{100}\right)$$

$$= P\left(\frac{-100}{100} \leq Z \leq \frac{100}{100}\right)$$

$$= P(-1 \leq Z \leq 1)$$

$$= \Phi_b - \Phi_a$$

$$= \Phi(1) - \Phi(-1)$$

$$= 0.84 - 0.15 = 0.69$$

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Example No 3:

Given: $\mu = 20$ $\sigma = 5$

i- between 20 and 30.

Sol. $P(20 \leq x \leq 30)$

$$= P\left(\frac{20-\mu}{\sigma} \leq \frac{x-\mu}{\sigma} \leq \frac{30-\mu}{\sigma}\right)$$

$$= P\left(\frac{20-20}{5} \leq Z \leq \frac{30-20}{5}\right)$$

$$= P(0 \leq Z \leq 2)$$

$$= \Phi(2) - \Phi(0)$$

$$= 0.97 - 0.50$$

$$= 0.47$$

ii- Less than 25

Sol. $P(x \leq 25)$

$$P\left(\frac{x-\mu}{\sigma} \leq \frac{25-\mu}{\sigma}\right)$$

$$P\left(Z \leq \frac{25-20}{5}\right)$$

$$P(Z \leq 1) = \Phi(1)$$

$$= 0.84134$$

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iii- More than 30

Sol. $P(x \geq 30)$

$$P\left(\frac{x-\mu}{\sigma} \geq \frac{30-\mu}{\sigma}\right)$$

$$P\left(Z \geq \frac{30-20}{5}\right)$$

$$P(Z \geq 2) = 1 - \Phi(2)$$

$$= 1 - 0.97725$$

$$= 0.02275$$

iv- Between 12 and 18.

Sol. $P(12 \leq x \leq 18)$

$$= P\left(\frac{12-\mu}{\sigma} \leq \frac{x-\mu}{\sigma} \leq \frac{18-\mu}{\sigma}\right)$$

$$= P\left(\frac{12-20}{5} \leq Z \leq \frac{18-20}{5}\right)$$

$$= P\left(-\frac{8}{5} \leq Z \leq -\frac{2}{5}\right)$$

$$= P(-1.6 \leq Z \leq -0.4)$$

$$= \Phi(-0.4) - \Phi(-1.6)$$

$$= 0.24458 - 0.05480$$

$$= 0.28978$$

v- Between 30 and 40

(Same method)

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Example No 02:

Given: $\mu = 25$
 $\sigma = 4$

i- More than 26 sales:

$$\begin{aligned} \text{Sol: } & P(X \geq 26) \\ &= P\left(\frac{X - \mu}{\sigma} \geq \frac{26 - 25}{4}\right) \\ &= P(Z \geq \frac{1}{4}) \end{aligned}$$

$$\begin{aligned} &= P(Z \geq 0.25) \\ &= 1 - \Phi(0.25) \\ &= 1 - 0.59871 \\ &= 0.40129 \end{aligned}$$

ii- Fewer than 20 sales:

$$\begin{aligned} &= P(X \leq 20) \\ &= P\left(\frac{X - \mu}{\sigma} \leq \frac{20 - 25}{4}\right) \\ &= P\left(Z \leq \frac{-5}{4}\right) \\ &= P(Z \leq -1.25) \\ &= \Phi(-1.25) \\ &= 0.10565 \end{aligned}$$

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iii) At least 25 or at most 25

$$\begin{aligned} &= P(25 \leq X \leq 25) \\ &= P\left(\frac{25 - \mu}{\sigma} \leq \frac{X - \mu}{\sigma} \leq \frac{25 - \mu}{\sigma}\right) \\ &= P\left(\frac{25 - 25}{4} \leq Z \leq \frac{25 - 25}{4}\right) \\ &= P(0 \leq Z \leq 0) \\ &= \Phi(0) - \Phi(0) \\ &= 0.50000 - 0.50000 \\ &= \boxed{0} \end{aligned}$$

Example No 06:

Given: ~~$\mu = 25$~~ $\mu = 15$

$X \sim N(15, 4)$ $X \sim N(\mu, \sigma^2)$

Here, $\mu = 15$, $\sigma^2 = 4$
 $\sigma = 2$

i) $P(X \leq 40)$

$$\begin{aligned} \text{Sol: } &= P(X \leq 40) \\ &= P\left(\frac{X - \mu}{\sigma} \leq \frac{40 - 15}{2}\right) \\ &= P\left(Z \leq \frac{40 - 15}{2}\right) \\ &= P(Z \leq 12.5) \\ &= \Phi(12.5) = 1.00 \end{aligned}$$

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ii) $P(X \geq 52)$

Sol. $= P(X \geq 52)$

$$= P\left(\frac{x - \mu}{\sigma} \geq \frac{52 - \mu}{\sigma}\right)$$

$$= P\left(Z \geq \frac{52 - 15}{2}\right)$$

$$= P(Z \geq 18.5)$$

$$= 1 - \Phi(18.5)$$

$$= 1 - 1$$

$$= 0$$

iii) $P(47 \leq x \leq 55)$

Sol. $= P(47 \leq x \leq 55)$

$$= P\left(\frac{47 - \mu}{\sigma} \leq \frac{x - \mu}{\sigma} \leq \frac{55 - \mu}{\sigma}\right)$$

$$= P\left(\frac{47 - 15}{2} \leq Z \leq \frac{55 - 15}{2}\right)$$

$$= P(16 \leq Z \leq 40)$$

$$= \Phi(40) - \Phi(16)$$

$$= 1 - 1$$

$$= 0$$

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Example NO 05:

Given: $\mu = 0.154$ inch

$$\sigma = 0.002 \text{ inch}$$

$$\text{Range} = 0.150 \text{ to } 0.155$$

(i) Accepted:

We find the probability that the thickness is between 0.150 and 0.155.

$$= P(0.150 \leq x \leq 0.155)$$

$$= P\left(\frac{0.150 - \mu}{\sigma} \leq x \leq \frac{0.155 - \mu}{\sigma}\right)$$

$$= P\left(\frac{0.150 - 0.154}{0.002} \leq Z \leq \frac{0.155 - 0.154}{0.002}\right)$$

$$= P(-2 \leq Z \leq 0.5)$$

$$= \Phi(0.5) - \Phi(-2)$$

$$= 0.69146 - 0.02275$$

$$= \boxed{0.66871}$$

(ii) Rejected:

$$P(\text{Rejected}) = 1 - P(\text{Accepted})$$

$$= 1 - 0.66871$$

$$= \boxed{0.33129}$$

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Example No 04:

Given: $\mu = 5 \text{ Feet } 10 \text{ inch} = 5(12) + 10 = 70 \text{ inch}$
 $\sigma = 2 \text{ inch}$

i) Freshmen taller than 6 Feet and 3 inch

$$= P(x \geq 75)$$

$$= P\left(\frac{x - \mu}{\sigma} \geq \frac{75 - 70}{2}\right)$$

$$= P\left(z \geq \frac{75 - 70}{2}\right)$$

$$= P(z \geq 2.5)$$

$$= 1 - \Phi(2.5)$$

$$= 1 - 0.99379$$

$$= \boxed{0.00621}$$

ii) Freshmen less than 5 Feet 7 inches:

(Same)

iii) Freshmen b/w 5 feet 8 and 5 feet 0 inches:

(same)

Example No 07:

(same)

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Properties of Normal Distribution:

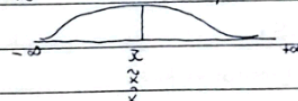
i- It is continuous

ii- It range from $-\infty$ to $+\infty$ ($-\infty \leq x \leq +\infty$)

iii- The parameters are μ (mean) and σ (S.D).

iv- The mean, median, and mode are equal in normal distribution.

v- It is bell-shaped distribution



vi- Total area under the normal curve is unity.